



ROLE RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9 Compute/Storage Senior Consultant — Success & Excel Runbook

Everything to walk in ready and excel as a VCF 9 compute/storage delivery consultant — the role boundary, the VCF 9 mental model, the workload-domain build lifecycle, the tool-by-tool reference, the troubleshooting playbook, the daily rhythm, and the interview-ready term corrections. Grounded in Broadcom TechDocs for VCF 9.0/9.1. No guessing — unverified points are flagged.

v1.0

VMware Cloud Foundation 9.0 / 9.1

© 2026 Virtual Control LLC — All Rights Reserved

INTERNAL

VCF 9 Compute/Storage Senior Consultant — Success & Excel Runbook

Prepared by: Virtual Control LLC **Date:** July 28, 2026 **Engagement:** VCF Compute/Storage Senior Consultant — World Wide Technology (WWT) / Broadcom VCF Practice, via Cloud Destinations **Baseline:** VMware Cloud Foundation 9.0 / 9.1 (verified against Broadcom TechDocs)

The one-sentence definition. A VCF Compute/Storage role means: **you build, integrate, validate, and troubleshoot the workload (compute) domains — the actual customer clusters — inside a VCF 9 environment.** You execute the architect's design with precision; you do not own the architecture. This is the tactical delivery tier, and it is where hands-on VCF 9 depth wins.

1. The Role Boundary — Architect Designs, You Execute + Validate

VCF has two big buckets. Know which one you live in.

Management Domain	Workload (Compute) Domains — your focus
SDDC Manager, management vCenter, management NSX, management vSAN, the fleet lifecycle plane	The customer-facing clusters where VMs and apps run, NSX segments live, vSAN policies apply
Deployed first, during fleet bring-up	Built <i>after</i> , on demand, as the customer scales horizontally
You integrate <i>with</i> it (register components, validate workflows)	You <i>build and validate</i> these — vCenter, NSX, vSAN ESA, clusters

Architect vs. you — the boundary, spelled out:

VCF Architect	VCF Compute/Storage Consultant (you)
Designs the environment; diagrams; network/storage/security models	Executes the architect's build plan
Chooses ESA vs OSA, host counts, cluster sizes, fault domains	Builds the workload domains and clusters to that spec
Writes the build plan; owns customer strategy	Deploys NSX, vSAN ESA, vCenters; validates integration points
—	Troubleshoots bring-up failures; documents as-built; Day-1/Day-2 support; KT to customer teams

In one line: *Architect = design. You = execution + validation. You make the design actually work.*

2. The VCF 9 Mental Model — What Changed From 4.x/5.x (READ THIS FIRST)

This is the single most important section. The compute work itself is what you already do; the *terminology and front-door changed* in VCF 9. Using 4.x/5.x muscle memory in front of a Broadcom PS lead is the fastest way to look dated. Correct it now.

The three muscle-memory corrections that matter most: 1. Front door moved to VCF Operations.

Workload-domain and many host/day-2 actions are now driven from **VCF Operations**, not "SDDC Manager" as the console. SDDC Manager still exists in 9.0 (lifecycle/binaries engine underneath) — but it is not the place you *start*. 2. **Images + Configuration Profiles replace Baselines + Host Profiles.** VCF 9.0+ supports **only vLCM image-based clusters** (baselines are gone). **Host Profiles are deprecated** → use **vSphere Configuration Profiles** (cluster-level JSON desired-config). 3. **The standalone Fleet appliance is 9.0-only.** In 9.1 it is removed and its role becomes **Fleet Lifecycle inside VCF Management Services**.

Concept	VCF 4.x / 5.x (old muscle memory)	VCF 9.0 / 9.1 (current — say this)
Host branding	"ESXi hosts"	"ESX" hosts (rebranded in 9.0 docs)
Free pool	"free pool"	"Unassigned Hosts" pool
Add hosts	Commission hosts (SDDC Mgr)	"Commission ESX Hosts" still exists in 9.0 ; hosts land in Unassigned Hosts ; drive from vSphere Client or SDDC Manager UI (9.1 <i>host-commission location: UNVERIFIED — confirm live</i>)
Create workload domain	SDDC Manager UI	VCF Operations : Inventory → Detailed View → VCF Instances → Add Workload Domain → Create New
Cluster lifecycle	Baselines / baseline groups (VUM)	vLCM images only (desired-state); baselines unsupported
Host config	Host Profiles	vSphere Configuration Profiles (JSON, cluster-level)
Deploy the platform	Cloud Builder	VCF Installer → deploys a VCF Fleet and VCF Instance(s)
Central console	vRealize Operations (separate)	VCF Operations — mandatory per-fleet console (monitoring, logs, licensing, Fleet Management)
Fleet lifecycle plane	(n/a)	9.0: standalone Fleet Management Appliance . 9.1: gone → VCF Management Services (Fleet Lifecycle + SDDC Lifecycle)
Brownfield onboarding	bring-up only	Import an existing vCenter to create a workload domain (supported 9.x path)

3. The Workload-Domain Build Lifecycle (Verified Flow)

The delivery arc, start to finish. Exact wizard labels can vary as Broadcom's doc set is mid-transition (some URLs still say "SDDC Manager" while the page body says VCF Operations) — treat **VCF Operations** as the authoritative action location.

3.1 Prepare & add hosts

1. **Validate host readiness** *before* touching VCF: firmware/BIOS, iDRAC/iLO, physical NIC/uplinks, **VLANs, MTU (jumbo where required), DNS (forward + reverse), NTP, certificates.**
2. **Commission ESX hosts** → they enter the **Unassigned Hosts** pool. Commission with the **target principal-storage type** (vSAN ESA/OSA, or other) that the destination cluster will use. Single-host or **JSON bulk**.

3.2 Create the VI workload domain (in VCF Operations)

3. **VCF Operations** → **Inventory** → **Detailed View** → **VCF Instances** → **Add Workload Domain** → **Create New** launches the **Workload Domain wizard**.
4. Wizard covers: **General/SSO** → **vCenter** → **cluster** → **cluster image (vLCM)** → **NSX Manager** → **storage** → **host selection** → **distributed switch** → **optional vSphere Supervisor** → **review**.
5. **Each workload domain gets its own dedicated vCenter.** A workload domain = **1 vCenter + an NSX Manager (shareable) + one or more vSphere clusters.**
6. *Brownfield option:* **import an existing vCenter** to form the workload domain instead of a clean deploy.

3.3 NSX integration — new or shared

7. During creation, **create a new NSX Manager instance or join an existing one.** Sharing NSX across domains is a first-class 9.x option — decide per the architect's design.

If you share an NSX instance across domains, these constraints bite (know them cold): - **One overlay transport zone per NSX instance** — every cluster on that NSX shares the same overlay TZ. - **IP coupling:** if the shared NSX was deployed IPv4-only, a new joining domain must use IPv4 for inter-component comms even if otherwise dual-stack. - **Lifecycle coupling:** you must **upgrade NSX + vCenter + ESX across all domains sharing that NSX together.** - Broadcom does **not** publish a single "default" (shared vs per-domain) — it's a **design decision.** Don't call one "the recommended default."

3.4 Configure storage (vSAN ESA / OSA)

8. Configure the cluster's **principal storage.** Both **vSAN ESA and OSA are valid principal storage in VCF 9.** - **ESA** — for high-performance / low-latency / high-IOPS; **single storage pool per host** (all devices contribute to capacity *and* performance, no cache/capacity tiering); **RAID-5/6 erasure coding** is the norm; **auto-policy management** sizes the default policy to cluster size/topology; hosts must be on the **vSAN ESA ReadyNode HCL.** - **OSA** — still fully supported; choose it when newer ESA-ready hardware isn't available. - **Stretched clusters** (ESA or OSA): deployed via **JSON spec + API**; in a workload domain the **management cluster must already be stretched first**; multi-rack ESA needs a default storage policy compatible with the rack/fault-domain count.

Verified-safe phrasing (don't overclaim): "ESA is the preferred architecture for new/performance clusters; OSA remains supported; both are valid principal storage." Do **not** say "ESA is the default."

3.5 Validate & hand off

9. **Validate cluster health** — vSAN health/compliance, NSX transport-node status and tunnels, vCenter registration in the fleet, LCM bundle applicability, DNS/NTP/cert propagation.
10. **Document as-built + Day-1/Day-2 runbooks; KT to the customer team.**

4. Tool-by-Tool Reference (Real VCF 9 Tools Only)

Strike `vdm`. There is **no `vdm` CLI** in VCF / vSphere / NSX / vSAN. (`vdmadmin` belongs to **VMware Horizon** — a separate EUC product.) Never reference it in a VCF compute/storage/NSX context.

Tool	What you use it for
VCF Operations	The front door: create/expand workload domains, inventory, monitoring, logs, licensing, Fleet Management (lifecycle, identity, certs, passwords, config, tags)
SDDC Manager (9.0)	Underlying lifecycle/binaries engine; holds bundles to deploy/patch domains. Present in 9.0; folded into VCF Management Services in 9.1
VCF Installer	Deploys a new VCF Fleet / VCF Instance (replaces Cloud Builder framing)
vSphere Client	Cluster config, VMkernel setup, Commission Hosts (Unassigned Hosts), vSAN ESA config, storage policies, Configuration Profiles , vLCM images
NSX Manager	Transport nodes, uplink/TEP profiles, transport zones, Tier-0/Tier-1, Edge deploy
VCF PowerCLI	Primary automation surface — 7000+ cmdlets across vSphere, VCF Operations, vSAN, NSX, Update Manager
VCF CLI	New first-party CLI for VCF context/plugin/resource management
VCF REST APIs / SDKs	SDDC Manager API, VCF Installer API, VCF SDK 9.0 (incl. vSAN Management SDK for Java); JSON-spec driven domain/stretch operations
Real-Time Metrics APIs	New Prometheus-compatible edge metrics across ESX/vCenter/vSAN/NSX
esxcli (incl. <code>esxcli</code> <code>vsan ...</code>)	Host/hypervisor-level storage, network, vSAN node ops
nsxcli	NSX CLI on managers/edges/transport nodes — transport/TEP/tunnel diagnostics
vsish / objtool	Low-level vSAN object/state inspection (advanced diagnostics)

5. The Troubleshooting Playbook (Where You Win)

Your differentiator is root-cause, not clicks. These are the real failure classes in workload-domain delivery — the pattern, the likely causes, and where to look.

5.1 Host commission / add failures

- **Likely causes:** wrong/absent **VLAN**, **MTU mismatch** (uplink not jumbo end-to-end), **DNS forward and reverse** missing, **NTP** not synced, **certificate** not trusted/propagated, wrong **principal-storage type** at commission.
- **Look:** SDDC Manager / VCF Operations task detail, host `esxccli network ip interface + vmknics` MTU, DNS both directions, `w32tm` /host NTP status.

5.2 NSX deployment failures

- **Likely causes:** Edge node bring-up errors, **TEP** misconfiguration, **transport-zone** mismatch, **uplink-profile** errors, **transport-node** ↔ **VDS** mismatch (a UUID/switch mismatch is a classic outage), overlay-TZ collisions when sharing NSX.
- **Look:** NSX Manager transport-node status + tunnels, `nsxccli` on the transport node, VDS UUID vs NSX's expected switch, MTU on the TEP VLAN.

5.3 vSAN ESA issues

- **Likely causes:** disk-group / storage-pool formation failure (non-ReadyNode or non-HCL device), **policy-compliance** errors (default policy vs fault-domain/rack count), performance anomalies, fault-domain misalignment.
- **Look:** vSAN Skyline Health, `esxccli vsan ...`, storage-policy compliance, ESA ReadyNode HCL status, disk claim state.

5.4 Workload-domain bring-up failures

- **Likely causes:** vCenter deploy failure, **NSX Manager registration** issue, **LCM bundle** incompatibility, API-workflow errors, stale inventory/MoRef references from prior redeloys.
- **Look:** the VCF Operations task chain, SDDC Manager logs, DNS/cert dependencies, bundle applicability, inventory consistency.

5.5 Integration / lifecycle drift

- **Likely causes:** SDDC Manager not recognizing a component, certificate mismatches, network dependency failures, lifecycle drift across shared-NSX domains.
- **Look:** fleet inventory state, cert trust chains, `version-sync` /inventory reconciliation, component registration.

Your banked, verified war stories map straight onto this — brownfield inventory desync recovery (Sync Inventory + guardrails), NSX transport-plane VDS-UUID rebuild, vpxd/cluster-store RCA, vSAN write-collapse from a stray tagged VMkernel, CPU co-stop on an over-wide appliance. Lead with these; they *are* this role's problem set.

6. The Daily Rhythm

Block	What you do
Morning	Review the architect's build plan; validate host readiness (iDRAC/iLO, firmware, VLANs, MTU, DNS, NTP); commission the day's hosts; validate cluster health
Midday	Deploy the workload domain / cluster; deploy vCenter + NSX Manager; configure transport nodes ; configure vSAN ESA ; apply storage policies; validate uplinks
Afternoon	Troubleshoot bring-up failures; validate LCM bundle applicability; run health checks ; document configs; update runbooks; KT with customer teams
End of day	Update the architect on progress; log issues + resolutions; stage next-day tasks

7. How to Excel (Not Just Deliver)

- **Be the one whose builds validate clean the first time.** Pre-flight readiness (VLAN/MTU/DNS/NTP/cert/HCL) *before* commissioning is what separates senior delivery from rework.
- **Root-cause, don't retry-blindly.** When a bring-up fails, read the task chain and name the cause. That reputation is your moat.
- **Document as you go.** As-built + runbook + Day-2 guidance handed off cleanly is what gets you invited back and referred.
- **Translate for the customer.** Explaining VCF/NSX/vSAN clearly to customer teams (KT) is a senior differentiator — you already do this well.
- **Respect the boundary.** Flag design concerns to the architect; don't silently redesign. Execution excellence + good escalation = trusted consultant.
- **Speak current VCF 9.** VCF Operations front door, images + config profiles, VCF Installer, ESA-or-OSA (not "ESA default"), 9.0-vs-9.1 fleet plane. Precision signals you're on today's product.
- **Bring your tooling.** Your health-check/validation utilities and RCA authoring prove an environment before and after handoff — offer them where engagement rules allow.

8. Interview / Positioning Cheat-Sheet

Say this: - "I build and validate workload domains — I execute the architect's design and make it work end to end." - "VCF 9 drives domain and host operations from VCF Operations; SDDC Manager is the lifecycle engine underneath in 9.0, and in 9.1 the Fleet plane consolidates into VCF Management Services." - "Clusters are vLCM image-based with vSphere Configuration Profiles for host config — baselines and host profiles are retired." - "ESA for performance clusters, OSA still supported — both are valid principal storage." - "I've root-caused the failures that stall builds: NSX transport-node/VDS mismatches, vSAN issues, brownfield inventory desync, cluster-store/vpxd faults."

Don't say: "vdm," "host profiles" (for config), "baselines," "commission in SDDC Manager as the console," "ESA is the default," "free pool" (it's **Unassigned Hosts**), "ESXi" where the 9.0 docs say **ESX**.

Don't overclaim a default where Broadcom frames a design choice: **NSX shared-vs-per-domain** and **ESA-vs-OSA** are *decisions*, not defaults.

9. Open Verification Items (confirm live on the engagement — no guessing)

- **9.1 host commissioning** — whether "Commission ESX Hosts" remains a discrete action in 9.1 or is absorbed into VCF Operations inventory. **UNVERIFIED** — check the live 9.1 host-management workflow.
- **Minimum host counts / domain-creation validation rules** in 9.x — not researched; confirm against the engagement's design and current docs.
- **Exact wizard labels** — the doc set is mid-transition (URL says "SDDC Manager," body says "VCF Operations"); confirm against the actual UI version on site.

10. Sources (Broadcom TechDocs — VCF 9.0/9.1)

- Commission Hosts — [.../9-0/building-your-private-cloud-infrastructure/host-management/commission-hosts.html](#)
- Deploy a VI Workload Domain — [.../9-0/building-your-private-cloud-infrastructure/working-with-workload-domains/deploy-a-vi-workload-domain-using-the-sddc-manager-ui.html](#)
- Working with Workload Domains (NSX shared/join) — [.../9-0/building-your-private-cloud-infrastructure/working-with-workload-domains.html](#)
- Import an existing vCenter — [.../working-with-workload-domains/import-an-existing-vcenter-to-create-a-workload-domain.html](#)
- Storage Models (ESA/OSA principal) — [.../9-0/design/vmware-cloud-foundation-concepts/storage-models.html](#)
- Stretch a Cluster — [.../9-0/building-your-private-cloud-infrastructure/stretching-clusters/stretch-a-cluster.html](#)
- vLCM baselines and images (baselines unsupported 9.0) — [.../vsphere/9-0/managing-host-and-cluster-lifecycle/about-vsphere-lifecycle-manager-new/vlcm-baselines-and-images.html](#)
- vSphere Configuration Profiles — [.../vsphere/9-0/managing-host-and-cluster-lifecycle/using-vsphere-config-profiles-to-manage-host-configuration-at-a-cluster-level.html](#)
- Fleet Management overview (9.0) — [.../9-0/overview-of-vmware-cloud-foundation-9/.../vcf-operations-overview/fleet-management.html](#)
- What's New — VCF Ops (9.1 Fleet appliance removed) — [.../9-1/release-notes/vmware-cloud-foundation-9-1-0-0-release-notes/what-s-new/whats-new-vcf-ops.html](#)
- VCF Management Services models (9.1) — [.../9-1/design/vmware-cloud-foundation-concepts/vcf-management-services-models.html](#)
- VCF CLI / API / SDK (9.0) — [.../9-0/release-notes/vmware-cloud-foundation-90-release-notes/platform-whats-new/whats-new-vcf-cli-api-sdk.html](#)

(All under <https://techdocs.broadcom.com/us/en/vmware-cis/>. Verified 2026-07-28.)

Document Information

Field	Value
Document ID	VC-RUNBOOK-VCF9-COMPUTE-CONSULTANT-20260728
Version	1.0
Issued	July 28, 2026

Field	Value
Author	Virtual Control LLC
Classification	Internal
Platform	VMware Cloud Foundation 9.0 / 9.1

Revision History

Version	Date	Notes
1.0	2026-07-28	Initial role runbook: role boundary, VCF 9 mental model + 4.x/5.x corrections, workload-domain build lifecycle, tool reference (vdm struck), troubleshooting playbook, daily rhythm, excel guidance, interview cheat-sheet. Grounded in Broadcom TechDocs (VCF 9.0/9.1); unverified points flagged.

Document End